07 — Determinism

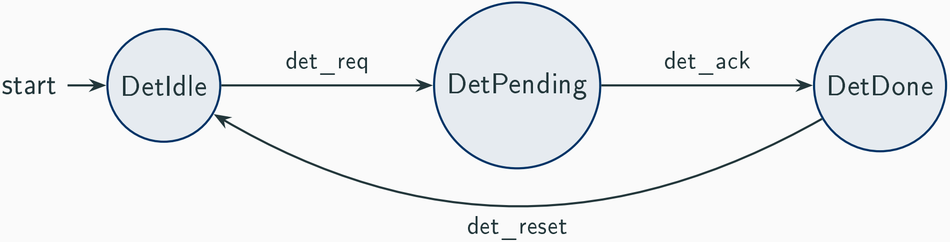
Deterministic vs Non-Deterministic Models

Mariano Cerrutti

Mununu Tutorial

Deterministic Protocol

In a **deterministic** automaton, each (state, label) pair has exactly one target:

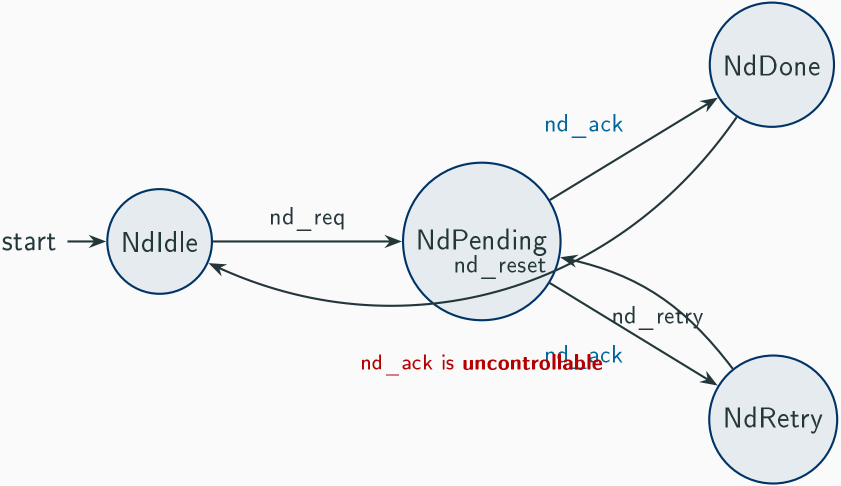


From **DetPending**, `det_ack` leads to *exactly one* state: **DetDone**.

No ambiguity — the next state is fully determined by the current state and the label.

Non-Deterministic Protocol

Now `nd_ack` from **NdPending** leads to two different states:



Impact on Verification: Box vs Diamond

Non-determinism changes the meaning of modalities:

$\square$ — **Box (universal):** "*All* successors satisfy φ"

$[\text{labels} = \{\texttt{nd\_ack}\}]$ *NdDone* is FALSE for NdPending
because one `nd_ack`-successor is NdRetry, not NdDone.

$\Diamond$ — **Diamond (existential):** "*Some* successor satisfies φ"

$\langle\text{labels} = \{\texttt{nd\_ack}\}\rangle$ *NdDone* is **TRUE** for NdPending
because one `nd_ack`-successor *is* NdDone.

In a **deterministic** model, box and diamond over a label agree (at most one successor).

Non-Deterministic Transitions in CTXDSL

```
automaton NondeterministicProtocol {
    controllable {
        label nd_req;
        label nd_reset;
        label nd_retry;
    }

    states {
        state NdIdle initial;
        state NdPending;
        state NdDone;
        state NdRetry;
    }

    transitions {
        transition NdIdle -> NdPending on label nd_req;
        // Non-deterministic: nd_ack leads to Done OR Retry
        transition NdPending -> NdDone on label nd_ack;
```

Formulas: Box vs Diamond

```
// NdDone reachability: only NdDone itself satisfies (1/4).
// From NdPending, nd_ack is nondeterministic AND uncontrollable:
// the environment triggers it and the outcome might be NdRetry,
// so the system cannot GUARANTEE reaching NdDone.
formula nd_done_reachable {
    over NondeterministicProtocol;
    body = mu X. (NdDone || <> X);
}

// Contrast: NdRetry is also only reachable from itself (1/4).
// Same reason --the nondeterministic nd_ack might go to NdDone.
formula nd_retry_reachable {
    over NondeterministicProtocol;
    body = mu X. (NdRetry || <> X);
}
```

- `nd_done_reachable` uses $\langle\ \rangle$ (diamond) — only **1/4** states satisfy, because nd_ack

Nondeterminism and the Skolem Paradigm

In Mununu's game-theoretic semantics, a player chooses a **label**, not a specific transition:

1. The controller (or environment) selects a label to fire.
2. If that label has multiple transitions from the current state, the outcome is **adversarial** — the worst case is assumed.
3. This is the **Skolem paradigm**: nondeterministic branching is resolved by the opponent.

Example from the protocol:

- `nd_done_reachable` = **1/4** states — from NdPending, diamond sees two nd_ack outcomes; since one leads to NdRetry (not NdDone), the adversary can force that branch.
- `nd_either_reachable` = **4/4** states — both nd_ack outcomes (NdDone and

The `< (ctrl = environment) >` Operator

The `< (ctrl = environment) >` diamond quantifies over **uncontrollable** transitions only:

```
        // Environment counterstrategy: env can prevent NdDone (3/4).
        formula nd_env_prevents_done {
            over NondeterministicProtocol;
            body = nu X. ((!NdDone) && < (ctrl = environment) > X);
        }
    }
```

$$\nu X.\,(\neg \mathrm{NdDone}) \;\wedge\; \langle \mathrm{ctrl} = \mathrm{environment} \rangle\; X$$

"The environment can keep the system away from NdDone forever."

- `_env_prevents_done`: Satisfied by **3/4** states (NdIdle, NdPending, NdRetry) — the environment triggers uncontrollable `nd_ack` and nondeterminism may steer to NdRetry indefinitely.

Takeaway

Concept	Meaning
Deterministic	Each (state, label) has at most one target. Fully predictable.
Non-deterministic	Same (state, label) can have multiple targets. Models uncertainty.
Box $[\,]\,\varphi$	Universal — *all* successors satisfy φ.
Diamond $\langle\,\rangle\,\varphi$	Existential — *some* successor satisfies φ.

Key insight: Non-determinism is not a bug — it models environments whose behavior we cannot predict. Choose box or diamond depending on whether you need *guarantees for all* or *existence of some* outcome.

Reference: Milner, R. (1989) *Communication and Concurrency*. Prentice Hall.